qtdecomp Function Test Results

Test Case 1: 1x1 Image

Input Matrix:

1.

Output Matrix:

1.

Test Case 2: eye(5,5)

Input Matrix:

1. 0. 0. 0. 0.

0. 1. 0. 0. 0.

0. 0. 1. 0. 0.

0. 0. 0. 1. 0.

0. 0. 0. 0. 1.

Output Matrix:

5. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

Test Case 3: eye(6,6)

Input Matrix:

1. 0. 0. 0. 0. 0.

0. 1. 0. 0. 0. 0.

0. 0. 1. 0. 0. 0.

0. 0. 0. 1. 0. 0.

0. 0. 0. 0. 1. 0.

0. 0. 0. 0. 0. 1.

Output Matrix:

3. 0. 0. 3. 0. 0.

0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0.

3. 0. 0. 3. 0. 0.

0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0.

<u>Test Case 4: eye(8,8)</u>

Input Matrix:

1. 0. 0. 0. 0. 0. 0. 0.

0. 1. 0. 0. 0. 0. 0. 0.

0. 0. 1. 0. 0. 0. 0. 0.

0. 0. 0. 1. 0. 0. 0. 0.

0. 0. 0. 0. 1. 0. 0. 0.

0. 0. 0. 0. 0. 1. 0. 0.

0. 0. 0. 0. 0. 0. 1. 0.

0. 0. 0. 0. 0. 0. 0. 1.

Output Matrix:

1. 1. 2. 0. 4. 0. 0. 0.

1. 1. 0. 0. 0. 0. 0. 0.

2. 0. 1. 1. 0. 0. 0. 0.

0. 0. 1. 1. 0. 0. 0. 0.

4.	0.	0.	0.	1.	1.	2.	0.
0.	0.	0.	0.	1.	1.	0.	0.
0.	0.	0.	0.	2.	0.	1.	1.
0.	0.	0.	0.	0.	0.	1.	1.

Test Case 5: Near-uniform Image

Input Matrix:

1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	3.	3.	1.	1.	1.
1.	1.	1.	3.	3.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.

Output Matrix:

2.	0.	2.	0.	2.	0.	2.	0.
0.	0.	0.	0.	0.	0.	0.	0.
2.	0.	1.	1.	1.	1.	2.	0.
0.	0.	1.	1.	1.	1.	0.	0.
2.	0.	1.	1.	1.	1.	2.	0.
0.	0.	1.	1.	1.	1.	0.	0.
2.	0.	2.	0.	2.	0.	2.	0.
0.	0.	0.	0.	0.	0.	0.	0.

Test Case 6: Zero Image

Input Matrix:

0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.

Output Matrix:

8.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.

<u>Test Case 7: Checkerboard</u>

Input Matrix:

0.	1.	0.	1.
1.	0.	1.	0.
0.	1.	0.	1.
1.	0.	1.	0.

Output Matrix:

1.	1.	1.	1.

1. 1. 1. 1.

1. 1. 1. 1.

1. 1. 1. 1.

<u>Test Case 8: Threshold 0</u>

Input Matrix:

1. 1. 1. 1.

1. 1. 1. 1.

1. 1. 2. 2.

1. 1. 2. 2.

Output Matrix:

2. 0. 2. 0.

0. 0. 0. 0.

2. 0. 2. 0.

0. 0. 0. 0.

<u>Test Case 9: Threshold 1</u>

Input Matrix:

1. 1. 1. 1.

1. 1. 1. 1.

1. 1. 2. 2.

1. 1. 2. 2.

Output Matrix:

4. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

<u>Test Case 10: mindim = 2 (strict behavior)</u>

Input Matrix:

1. 1. 2. 2. 3. 3. 4. 4.

1. 1. 2. 2. 3. 3. 4. 4.

5. 5. 6. 6. 7. 7. 8. 8.

5. 5. 6. 6. 7. 7. 8. 8.

9. 9. 10. 10. 11. 11. 12. 12.

9. 9. 10. 10. 11. 11. 12. 12.

13. 13. 14. 14. 15. 15. 16. 16.

13. 13. 14. 14. 15. 15. 16. 16.

Output Matrix:

4. 0. 0. 0. 4. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

4. 0. 0. 0. 4. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

Block uniqueness check (should NOT contain 1x1 splitting artifacts): 0.

<u>Test Case 11: mindim=2 maxdim=4</u>

Input Matrix:

1. 1. 1. 1. 2. 2. 2. 2.

1.	1.	1.	1.	2.	2.	2.	2.
1.	1.	1.	1.	2.	2.	2.	2.
1.	1.	1.	1.	2.	2.	2.	2.
3.	3.	3.	3.	4.	4.	4.	4.
3.	3.	3.	3.	4.	4.	4.	4.
3.	3.	3.	3.	4.	4.	4.	4.
3.	3.	3.	3.	4.	4.	4.	4.

Output Matrix:

4.	0.	0.	0.	4.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
4.	0.	0.	0.	4.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.

<u>Test Case 12: Octave Regression A</u>

Input Matrix:

1.	4.	2.	5.	54.	55.	61.	62.
3.	6.	3.	1.	58.	53.	67.	65.
3.	6.	3.	1.	58.	53.	67.	65.
3.	6.	3.	1.	58.	53.	67.	65.
23.	42.	42.	42.	99.	99.	99.	99.
27.	42.	42.	42.	99.	99.	99.	99.
23.	22.	26.	25.	99.	99.	99.	99.
22.	22.	24.	22.	99.	99.	99.	99.

Output Matrix:

1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	2.	0.	4.	0.	0.	0.
1.	1.	0.	0.	0.	0.	0.	0.
1.	1.	1.	1.	0.	0.	0.	0.
1.	1.	1.	1.	0.	0.	0.	0.

Test Case 13: A threshold=5

Input Matrix:

1.	4.	2.	5.	54.	55.	61.	62.
3.	6.	3.	1.	58.	53.	67.	65.
3.	6.	3.	1.	58.	53.	67.	65.
3.	6.	3.	1.	58.	53.	67.	65.
23.	42.	42.	42.	99.	99.	99.	99.
27.	42.	42.	42.	99.	99.	99.	99.
23.	22.	26.	25.	99.	99.	99.	99.
22.	22.	24.	22.	99.	99.	99.	99.

Output Matrix:

4.	0.	0.	0.	2.	0.	1.	1.
0.	0.	0.	0.	0.	0.	1.	1.
0.	0.	0.	0.	2.	0.	2.	0.
0.	0.	0.	0.	0.	0.	0.	0.
1.	1.	2.	0.	4.	0.	0.	0.
1.	1.	0.	0.	0.	0.	0.	0.

2. 0. 2. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

Test Case 14: A threshold=10

Input Matrix:

1. 4. 2. 5. 54. 55. 61. 62.

3. 6. 3. 1. 58. 53. 67. 65.

3. 6. 3. 1. 58. 53. 67. 65.

3. 6. 3. 1. 58. 53. 67. 65.

23. 42. 42. 42. 99. 99. 99. 99.

27. 42. 42. 42. 99. 99. 99. 99.

23. 22. 26. 25. 99. 99. 99. 99.

22. 22. 24. 22. 99. 99. 99. 99.

Output Matrix:

4. 0. 0. 0. 2. 0. 2. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 2. 0. 2. 0.

0. 0. 0. 0. 0. 0. 0. 0.

1. 1. 2. 0. 4. 0. 0. 0.

1. 1. 0. 0. 0. 0. 0. 0.

2. 0. 2. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

Test Case 15: Function Handle

Input Matrix:

1. 4. 2. 5. 54. 55. 61. 62.

3. 6. 3. 1. 58. 53. 67. 65.

3.	6.	3.	1.	58.	53.	67.	65.
3.	6.	3.	1.	58.	53.	67.	65.
23.	42.	42.	42.	99.	99.	99.	99.
27.	42.	42.	42.	99.	99.	99.	99.
23.	22.	26.	25.	99.	99.	99.	99.
22.	22.	24.	22.	99.	99.	99.	99.

Output Matrix:

1.	1.	1.	1.	4.	0.	0.	0.
1.	1.	1.	1.	0.	0.	0.	0.
1.	1.	1.	1.	0.	0.	0.	0.
1.	1.	1.	1.	0.	0.	0.	0.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.
1.	1.	1.	1.	1.	1.	1.	1.

Test Case 16: Non-square image

Input Matrix:

0.5667211	0.5595937	0.5465335	0.5900573	0.1157417
0.5711639	0.124934	0.9885408	0.3096467	0.6117004
0.816011	0.7279222	0.7395657	0.2552206	0.6783956
0.0568928	0.2677766	0.0037173	0.6251879	0.3320095

Error: "qtdecomp: I should be square."

Test Case 17: Invalid dimensions

Input Matrix:

1. 0. 0. 0. 0. 0. 0. 0.

0. 1. 0. 0. 0. 0. 0. 0.

0. 0. 1. 0. 0. 0. 0. 0.

0. 0. 0. 1. 0. 0. 0. 0.

0. 0. 0. 0. 1. 0. 0. 0.

0. 0. 0. 0. 0. 1. 0. 0.

0. 0. 0. 0. 0. 0. 1. 0.

0. 0. 0. 0. 0. 0. 0. 1.

Error: "qtdecomp: mindim must be smaller than maxdim."

<u>Test Case 18: Large Uniform Image</u>

Input Matrix:

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

1. 1. 1. 1. 1. 1. 1. 1.

Output Matrix:

64. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

0. 0. 0. 0. 0. 0. 0. 0.

<u>Test Case 19: uint8 image</u>

Input Matrix:

10 10 10 10

10 10 10 10

10 10 10 10

10 10 10 10

Output Matrix:

4. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

0. 0. 0. 0.

<u>Test Case 20: Sparse Output</u>

Input Matrix:

1. 1. 2. 2.

1. 1. 2. 2.

3. 3. 4. 4.

3. 3. 4. 4.

Type of Matrix: "sparse"

Output Matrix:

2. 0. 2. 0.

0. 0. 0. 0.

2. 0. 2. 0.

0. 0. 0. 0.